



Checkpoint 2: first experiments/results

Activity: Checkpoint 2: Discuss an experiment + analysis

Course: 2526 Natural Computing (KW3 V)

Name: Robert Blaauwendraad

Methodology and experimental design	Insufficient	Sufficient	Good	Excellent
(*) Approach appropriate & justified?	Approach is not suitable to address goals, flawed or lacks justification.	The methodology is adequate; flaws are minor. Most choices are justified. ✓	Good methodology which may have only minor flaws and is well-justified.	Good methodology that is very well justified and has (almost) no flaws.
(*) Scope of methodological work (can be planned work)	Too similar to assignments or existing code without further exploration.	Sufficient exploration of the method and/or experiments.	Methods and expts are extensive with minor flaws, or of high quality. ✓	Extensively developed methods and/or experiments that are high quality.
(*) Quality of experimental design	Experimental design has major flaws (e.g., missing baselines, etc.).	Experiments are basic but informative, any flaws are minor. ✓	Experimental design is good but could be extended for generalizability.	Experiments are well-designed to address the questions and have no flaws.
Criterion Feedback Check PDF notes. Do provide more details on how the experimental design was made.				
(*) Correctness & understanding of the described methods.	Significant factual errors reveal insufficient understanding of the subject matter.	Explanations show basic grasp of the methodologies and use cases, but may have minor flaws. ✓	Explanations show a solid grasp of the methodology and its use cases.	The explanations show an in-depth understanding of the subject matter.
Analysis of results	Insufficient	Sufficient	Good	Excellent
(*) Quality of quantitative analysis of obtained results (not necessarily all the planned results, but at least one experiment)	Analysis is shallow, inaccurate, or lacks depth and fails to examine data critically.	Analysis with some relevant metrics, but there may still be room for further exploration.	The analysis thoroughly examines the obtained data with suitable metrics. ✓	The analysis goes beyond surface-level, offering insightful and robust interpretations of the data shown.

Analysis of results	Insufficient	Sufficient	Good	Excellent
(*) Quality of statistical analysis (can also be a plan for further statistical analysis)	Statistical analysis is necessary given the questions asked/conclusions drawn, but absent or strongly flawed.	Statistical analysis is either absent but not strictly needed for the problem (although it could have strengthened results), OR it is present and adequate for the goals, but could be improved. ✓	Statistical analysis is present & correct where necessary. There may still be room for extending it.	Statistical analysis is thoughtfully used where informative and correct.
(*) Quality of data visualization	Visualizations are absent/not suitable for the type of data, failing to show key patterns in the data.	The visualization communicates insights but may not optimally show relevant patterns. ✓	The visualization effectively communicates insights and relevant patterns.	Visualizations are tailored to highlight key points, helping the reader see important patterns.

Criterion Feedback

See PDF comments. You need to provide more details on how to read the results.

Conclusion, discussion, and interpretation	Insufficient	Sufficient	Good	Excellent
(*) Validity (interpretation of results so far)	Conclusions fail to match presented results, are inconsistent or have unsupported claims. ✓	Conclusions generally backed up by evidence. If claims are not fully supported, these are not key claims.	Conclusions accurately reflect results and consider key alternative explanations appropriately.	Conclusions rigorously match results with comprehensively analyzed alternative explanations.

Criterion Feedback

Missing. There are not yet conclusions in the current report.

Overall Score

Good to go!

Initial experiment(s) have a good design and are properly analyzed and interpreted. There is a clear plan for follow-up.

OK, but with caveats ✓

Initial experiment(s) are present with a mostly proper design / analysis / interpretation. There might be some (minor) flaws that can reasonably be addressed.

Same for the follow-up plan.

Danger zone

There are no initial experiments yet, or there are important flaws in terms of experiment design, analysis, interpretation and/or plan for follow-up.